2. **DPCM (Differential Pulse Code Modulation)**

- DPCM: a procedure of converting an analog signal into a digital signal in which an analog signal is sampled and then the difference between the actual sample value and its predicted value (predicted value is based on previous sample or samples) is quantized and then encoded forming a digital value.

- DPCM : code words represent differences between samples,

  unlike

- PCM : where code words represented a sample value

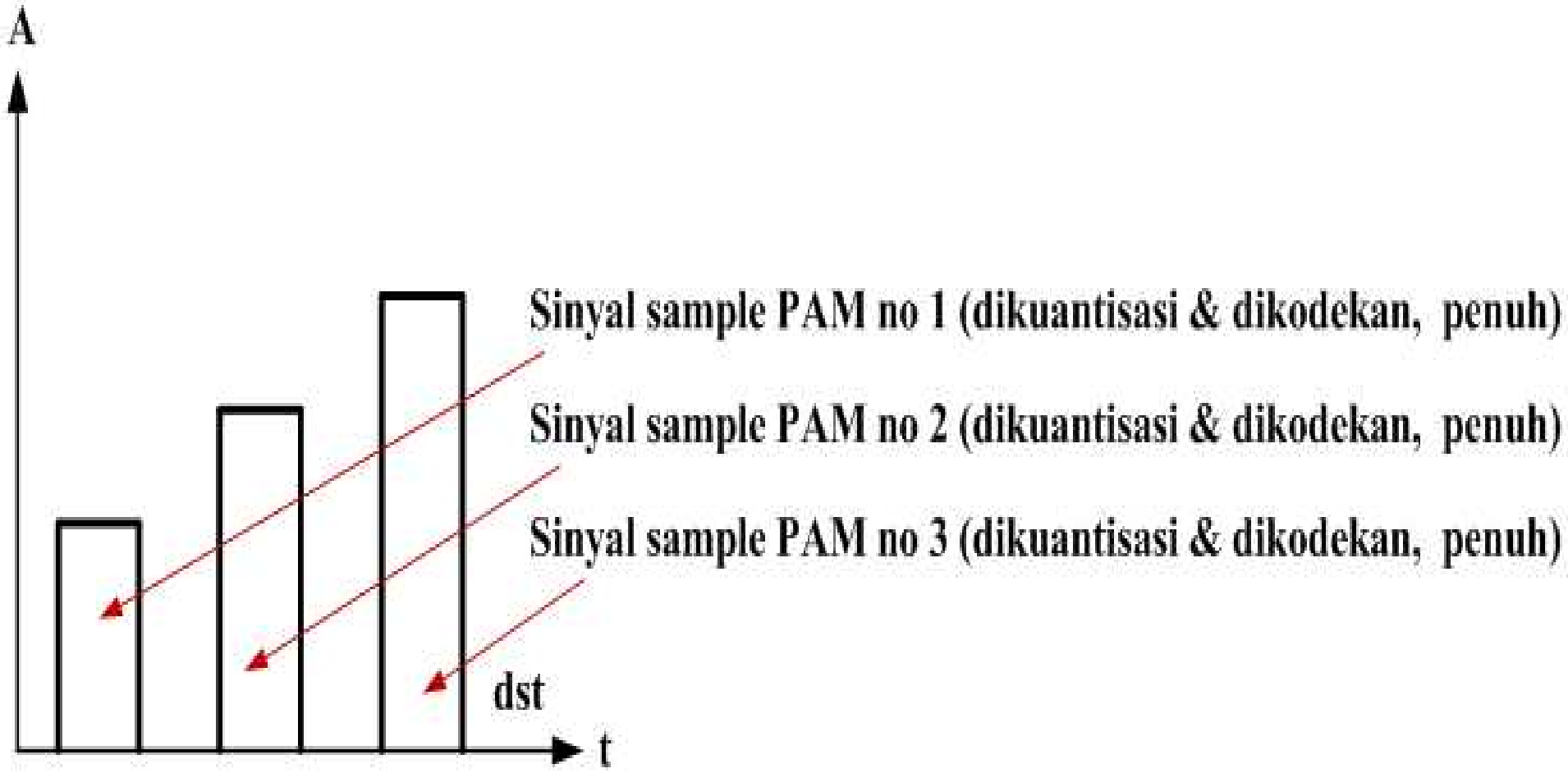
- Basic concept of DPCM : coding a difference, is based on the fact that most source signals show significant correlation between successive samples so encoding uses redundancy in sample values which implies lower bit rate.

- Realization of basic concept : is based on a technique in which we have to predict current sample value based upon previous samples (or sample) and we have to encode the difference between actual value of sample and predicted value (the difference between samples can be interpreted as prediction error).

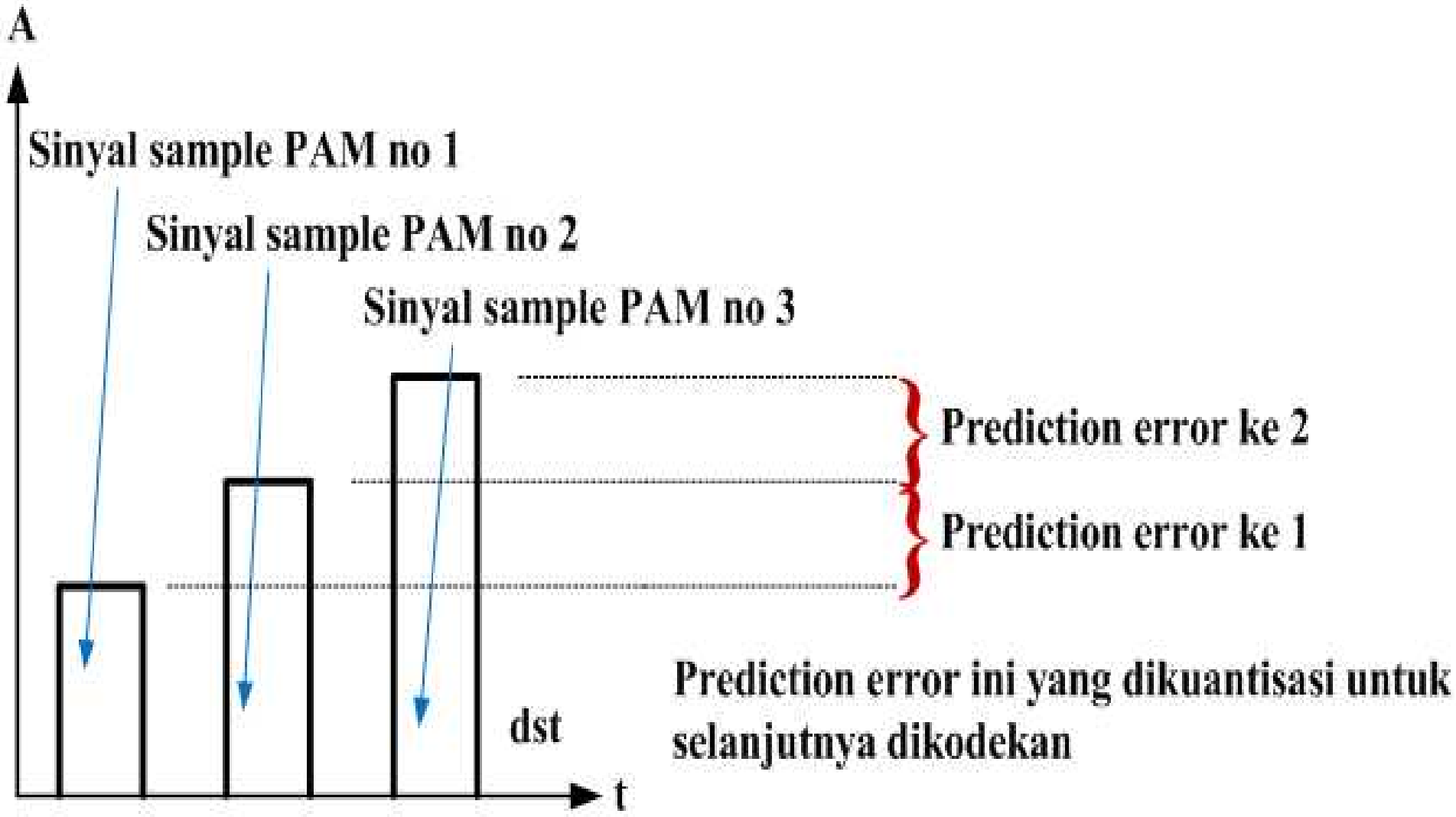
  Because it's necessary to predict sample value DPCM is form of predictive coding.

DPCM : Mengkodekan Perbedaan dua amplitudo sinyal sample PAM yang berdekatan, yaitu sinyal sample sebelumnya dengan sinyal sample berikutnya (yang disebut " prediction error), dikuantisasi selanjutnya dikodekan menjadi sinyal digital (code word) dan ditransmisikan.

- Perkiraan harga sinyal sample berikutnya dibuat berdasarkan atas harga sinyal sample terdahulu.

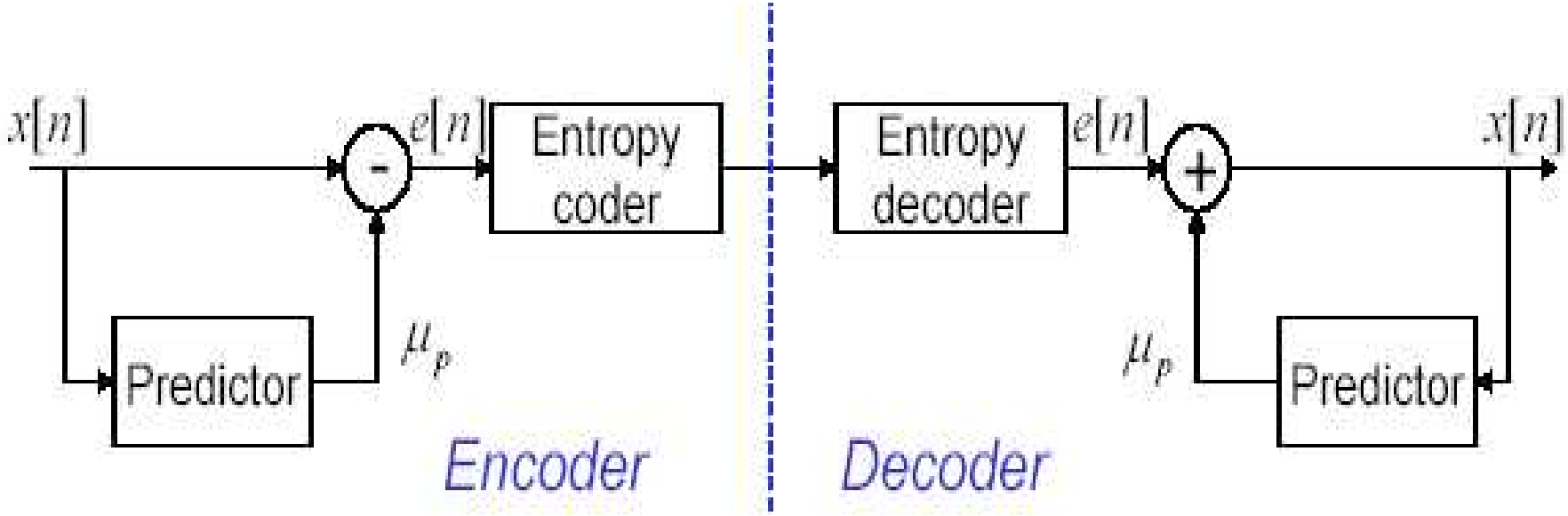


PCM : code words represented a sample value.



DPCM : code words represent differences between samples

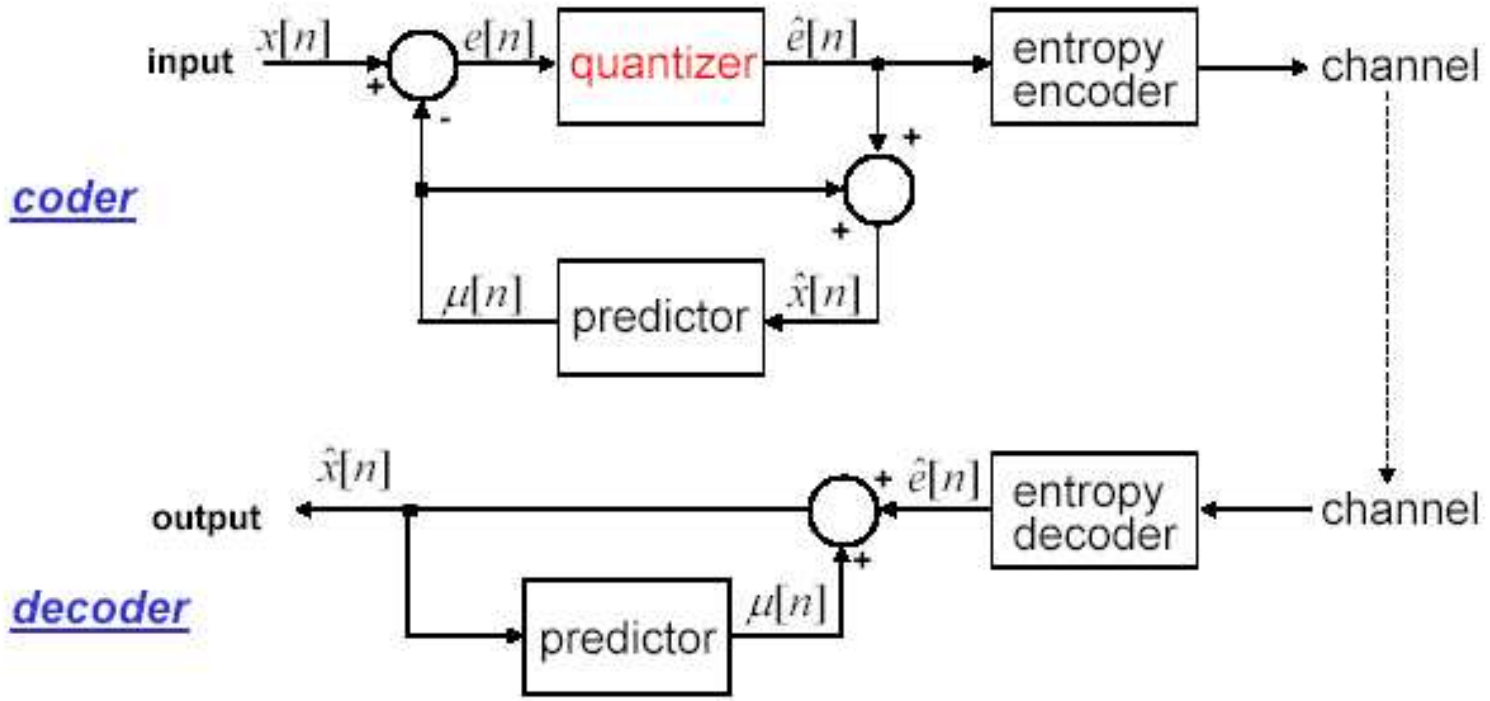
Reduksi Redundancy dg Prediction pd Lossless Coding



Principle

- *x[n]* adalah sampel saat ini dari sinyal original
- μ_p adalah prediksi utk *x[n]* dikalkulasi dari sampel sebelumnya
- *e[n]* adalah prediction error, yg secara signifikan mengurangi statistical dependencies antar sampel berdekatan
- Penerima dp merekonstruksi *x[n]* tanpa loss

Prinsip DPCM



Prediction error

$$e = x - \mu$$

Reconstruction

$$\hat{x} = \hat{e} + \mu$$

Reconstruction error = quantization error

$$q = e - \hat{e} = x - \hat{x}$$

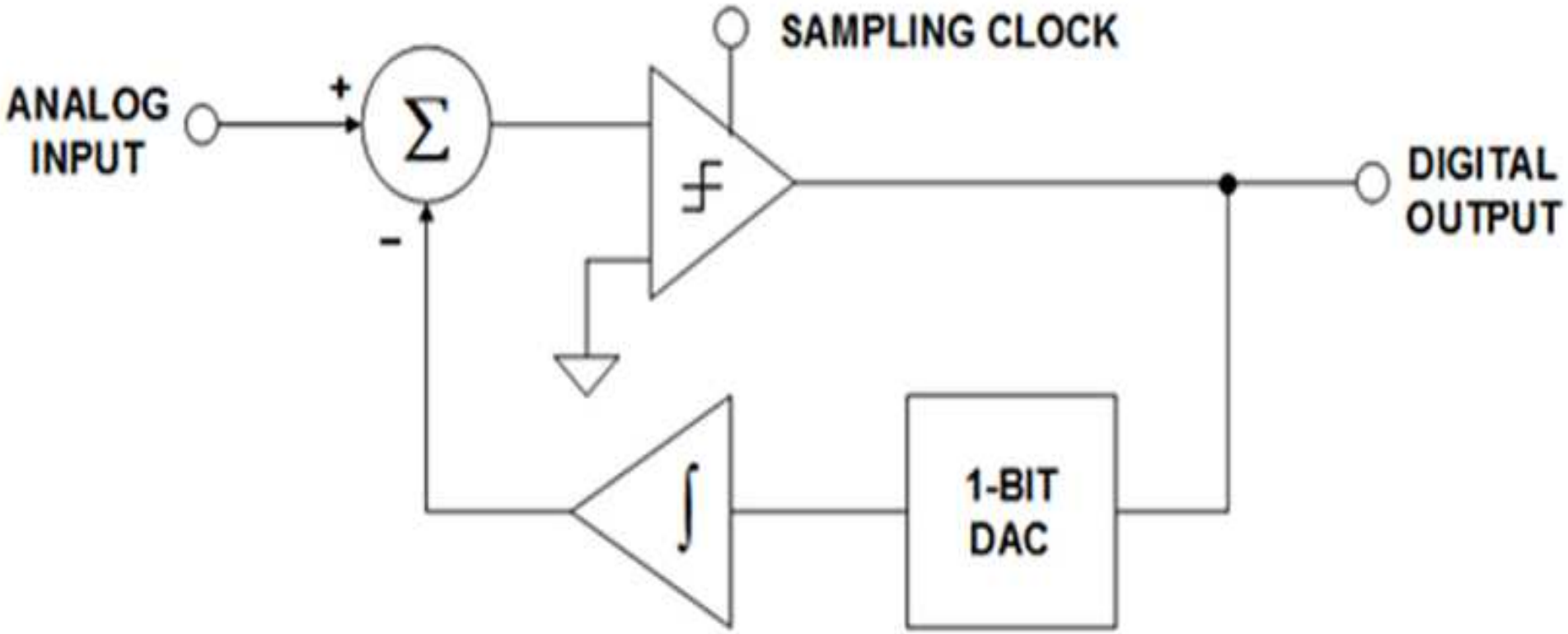
3. Modulasi Delta

- **D**elta **m**odulation (DM)is a subclass) of differential pulse code modulation (DPCM).
- It can be viewed as a simplified variant of DPCM, in which 1-bit quantizer is used with the fixed first order predictor, and was developed for voice telephony applications.

Principle of DM :

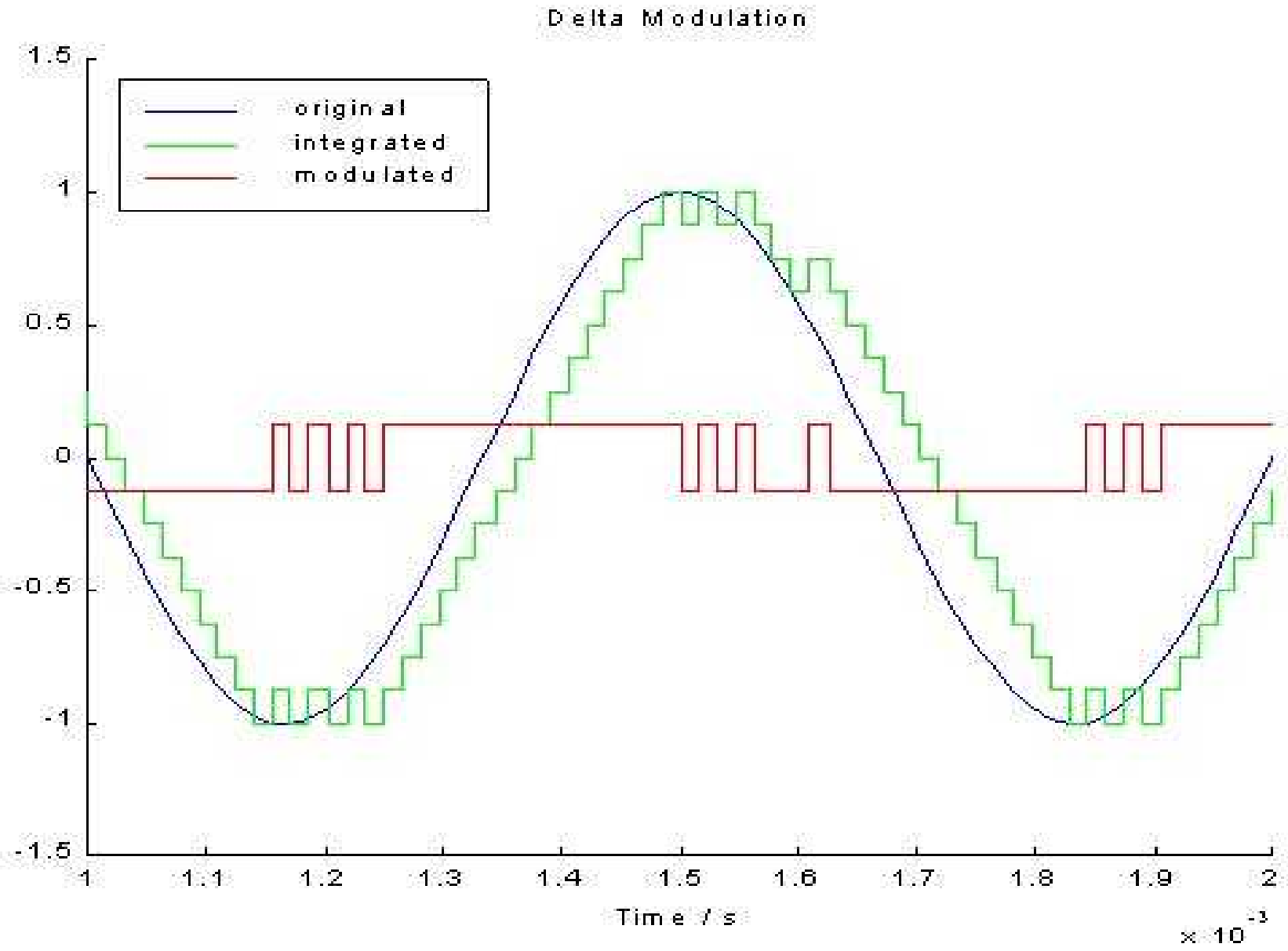
- Transmits information only to indicate whether the analog signal that is being encoded goes up or goes down.
- The Encoder Outputs are highs or lows that “instruct” whether to go up or down, respectively.
- DM output is 0 if waveform falls in value.
- DM output is 1 represents rise in value.
- Each bit indicates direction in which signal is changing (not how much), i.e.
- DM codes the direction of differences in signal amplitude instead of the value of difference (DPCM).

- The analog signal is quantized by a one-bit ADC (a comparator implemented as a comparator).
- The comparator output is converted back to an analog signal with a 1-bit DAC, and subtracted from the input after passing through an integrator.
- The shape of the analog signal is transmitted as follows:
  - a "1“ indicates that a positive excursion has occurred since the last sample,
  - and a "0" indicates that a negative excursion has occurred since the last sample.

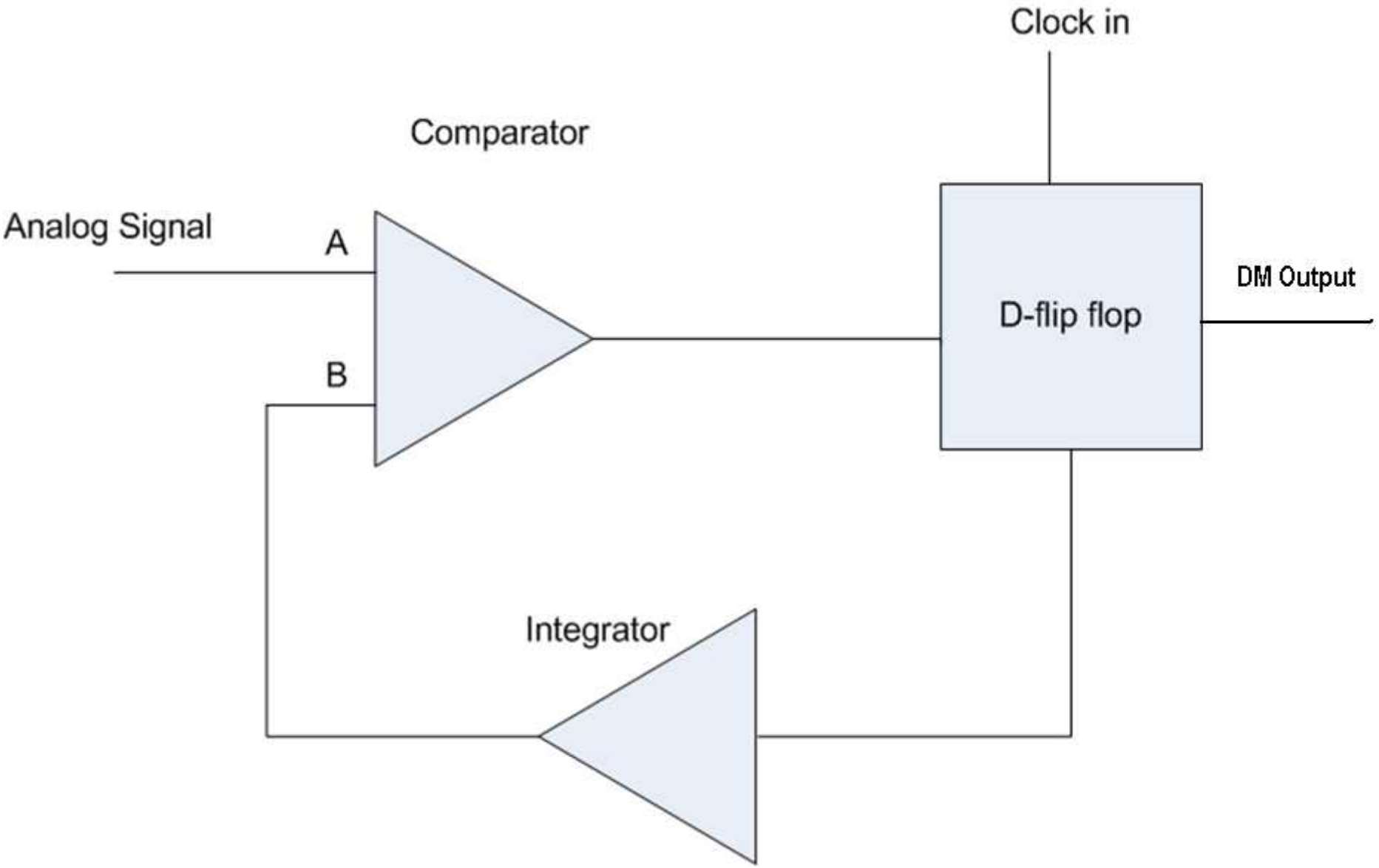


DELTA MODULATION

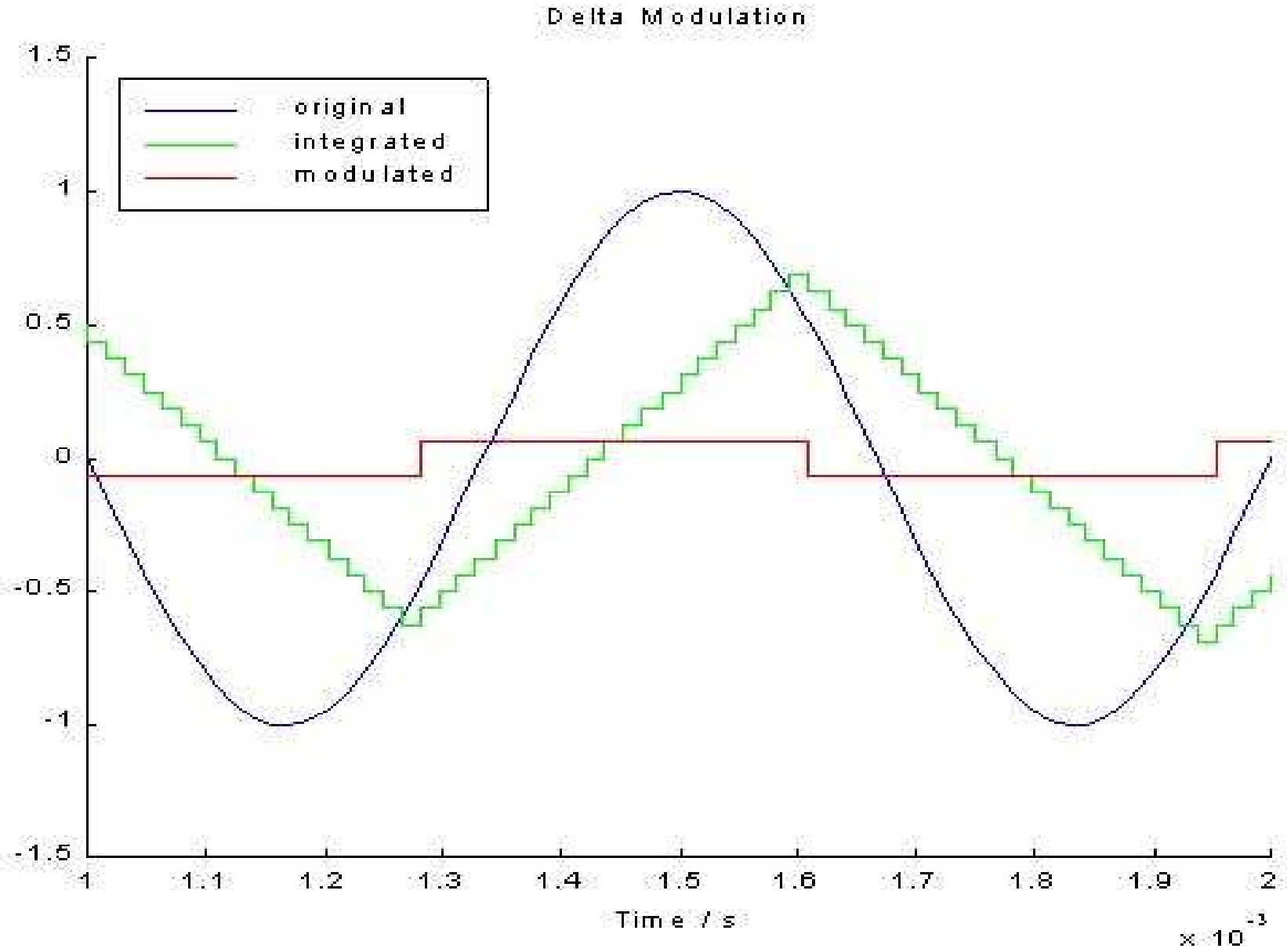
Resulting Waveforms

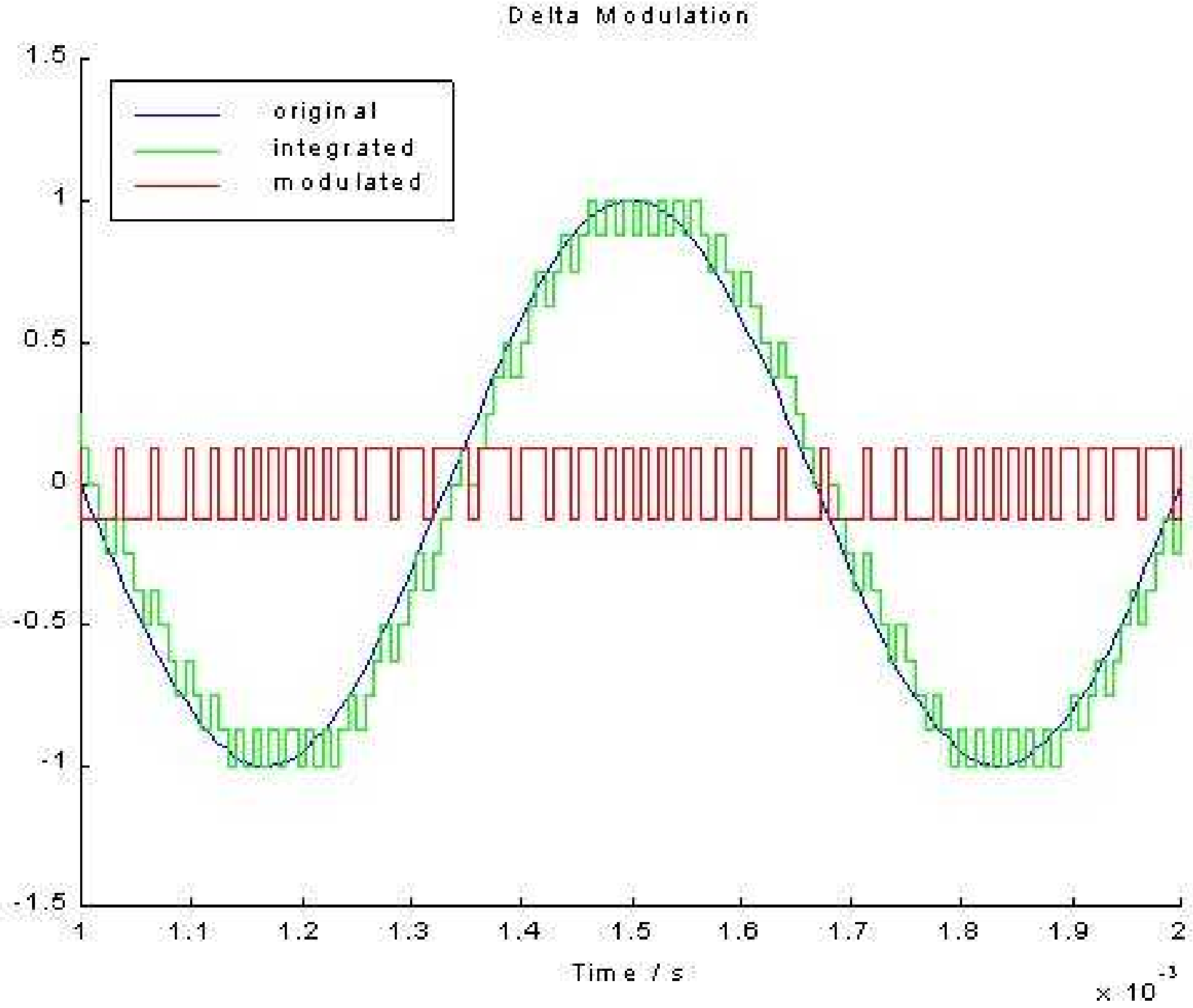


Simple Implementation of a DM system

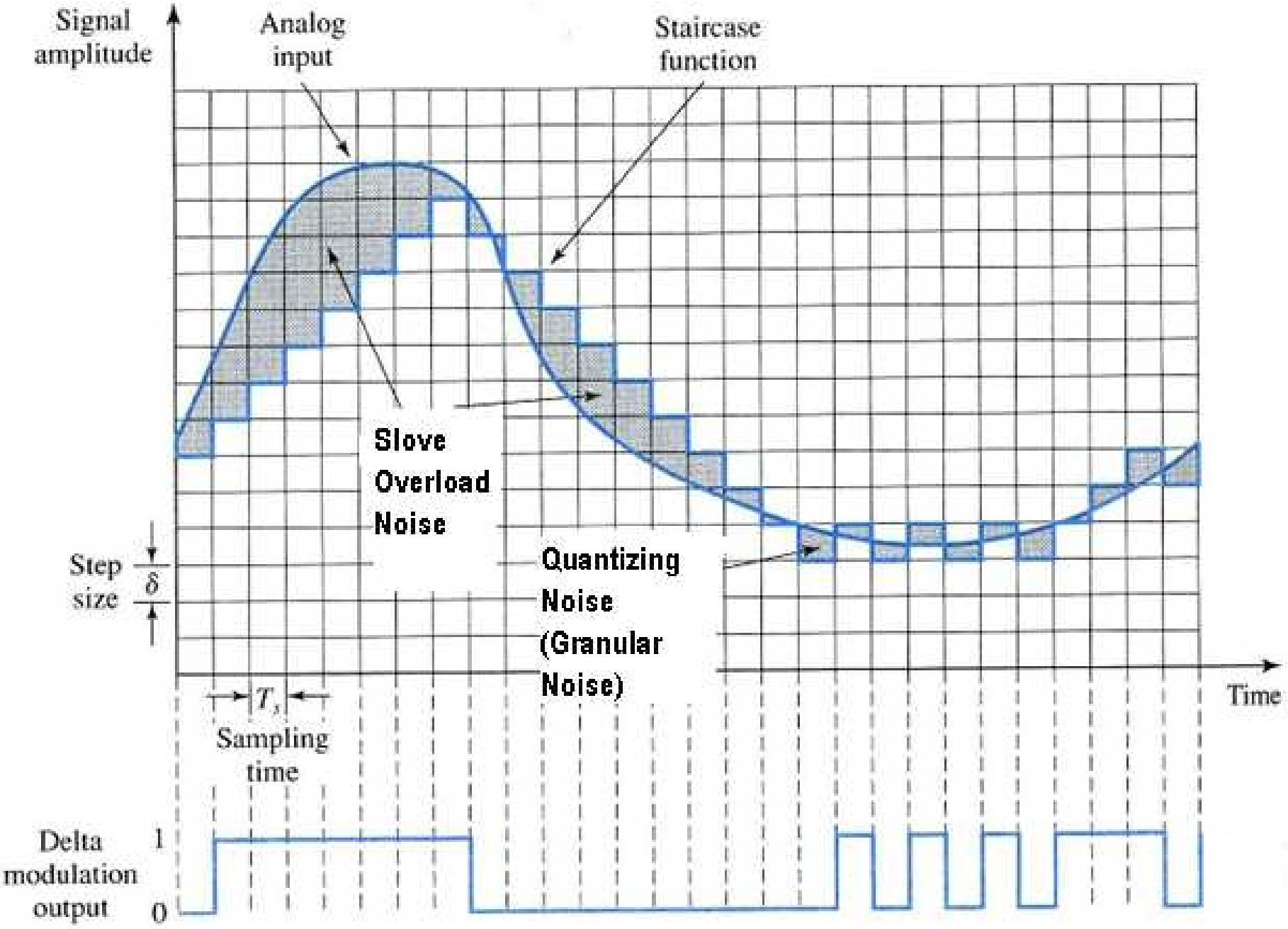


Limitations of the system




Delta Modulation
original
integrated
modulated
1.5
1
0.5
0
-0.5
-1
-1.5
1
1.1
1.2
1.3
1.4
1.5
1.6
1.7
1.8
1.9
2
Time / s
x 10^-3

- The codec determines if the waveform has risen or dropped one delta step.
- Transmit a 1 or 0 accordingly.
- Delta Modulation requires only 1 bit sample.
- Delta modulation is faster than PCM.
- ***Slope overload noise***: waveform ***rises or drops too quickly***, the codec not able to keep up with the change.
- Constant waveform: A ***non-changing waveform*** generates a pattern ***of 01010101.***.. thus generating ***quantising noise*** atau ***Granular noise***.


Signal amplitude
Analog input
Staircase function
Slove Overload Noise
Quantizing Noise (Granular Noise)
Step size
δ
Time
T_s
Sampling time
Delta modulation output
1
0